

Task-7HD

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0.1 Deakin University

0.2 SIG731- Data Wrangling

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0.3 Task-7HD

In this assignment task, we will explore the National Health and Nutrition Examination Center (NHANES) survey datasets for the years 2021-2023 and 2017-2020. These datasets contain information on Demographics, Examination, Dietary, Laboratory, and Questionnaire data, which we will use to perform data mining and derive meaningful insights through data analysis.

Our objective is to demonstrate how to load these datasets into Pandas DataFrames, merge them into a single dataset, and extract meaningful information to create visualizations and explain insights. This task will help you understand key data analysis processes, including data loading, data merging, identifying relevant data, visualization, and insight explanation.

0.3.1 Datasets Description

2021-2023

- **Demographics:** Demographics
- **Examination:** Body Measures
- **Dietary:** individual Foods, First Day
- **Laboratory:** alpha-1-Acid Glycoprotein
- **Questionnaire:** Alcohol Use

2017-2020

- **Demographics:** Demographics
- **Examination:** Body Measures
- **Dietary:** individual Foods, First Day
- **Laboratory:** alpha-1-Acid Glycoprotein
- **Questionnaire:** Alcohol Use

```
[1]: import pandas as pd
import numpy as np
import seaborn as sns
```

```

import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import requests
import pyreadstat
import io
from io import BytesIO

#Python warning ingorning
import warnings
warnings.filterwarnings('ignore')
pd.set_option("display.notebook_repr_html", False)

```

0.4 Selecting Nhmes URL fo finding dataset

0.4.1 (2021-2023) National Helath and Nutrition Examination Center (Nhames) survery dataset URL

```

[2]: #(2021-2023) URL
#August 2021-August 2023 Demographics Data - Continuous NHANES
url_demo_2021_2023 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2021/
↳DataFiles/DEMO_L.xpt"

# Examination Data: Body Measures
url_bmx_2021_2023 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2021/
↳DataFiles/BMX_L.xpt"

#Dietary Interview - Individual Foods, First Day
url_dr1review_2021_2023 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2021/
↳DataFiles/DR1IFF_L.xpt"

#Laboratory Data: alpha-1-Acid Glycoprotein
url_acid_glycoprotin_2021_2023 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/
↳2021/DataFiles/AGP_L.xpt"

#Questionnaire Data: Alcohol Use
url_alc_2021_2023 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2021/
↳DataFiles/ALQ_L.xpt"

```

2021-2023 Data

- August 2021-August 2023 Demographics Data - Continuous NHANES
- Examination Data: Body Measures
- Dietary Interview - Individual Foods, First Day
- Laboratory Data: alpha-1-Acid Glycoprotein
- Questionnaire Data: Alcohol Use

0.4.2 (2017-2020) National Health and Nutrition Examination Center (NHANES) survey dataset URL

```
[3]: #(2017-2020) URL
#August 2017-March 2020 Demographics Data - Continuous NHANES
url_demo_2017_2020 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2017/
↳DataFiles/P_DEMO.xpt"

# Examination Data: Body Measures
url_bmx_2017_2020 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2017/
↳DataFiles/P_BMX.xpt"

#Dietary Interview - Individual Foods, First Day
url_dr1review_2017_2020 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2017/
↳DataFiles/P_DR1IFF.xpt"

#Laboratory Data: alpha-1-Acid Glycoprotein
url_acid_glycoprotein_2017_2020 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/
↳2017/DataFiles/P_SSAGP.xpt"

#Questionnaire Data: Alcohol Use
url_alc_2017_2020 = "https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2017/
↳DataFiles/P_ALQ.xpt"
```

2017-2020 data

- August 2017-March 2020 Demographics Data - Continuous NHANES
- Examination Data: Body Measures
- Dietary Interview - Individual Foods, First Day
- Laboratory Data: alpha-1-Acid Glycoprotein
- Questionnaire Data: Alcohol Use

0.5 Downloading and Extracting Data

```
[4]: ## Data Extraction Download method and return the Pandas dataframe
def nhanes_data_extraction(url):
    response = requests.get(url)
    # Save the file locally
    with open("temp_data.xpt", "wb") as f:
        f.write(response.content)
    # Read the .XPT file using pyreadstat
    df, meta = pyreadstat.read_xport("temp_data.xpt", encoding='latin1')
    # Display the first few rows
    return df
```

- Define nhanes_data_extraction() method this method will Download data specify url
- write to local disk "temp_data.xpt" file
- Read the .XPT file using pyreadstat as pyreadstat package has to install !pip install pyreadstat

- Return Panda dataframe download file

0.5.1 Extracting data 2021-2023 National Helath and Nutrition Examination Center (Nhanes) in padas dataframe

```
[5]: # Demographics
df_demo_2021_2023 = nhanes_data_extraction(url_demo_2021_2023)

# Body Measures
df_bmx_2021_2023 = nhanes_data_extraction(url_bmx_2021_2023)

# Diatary data review
df_dr1review_2021_2023 = nhanes_data_extraction(url_dr1review_2021_2023)

#Laboratory Data: alpha-1-Acid Glycoprotein
df_acid_glycoprotin_2021_2023 = □
↳nhanes_data_extraction(url_acid_glycoprotin_2021_2023)

# Questionnaire Data: Alcohol Use data load
df_alc_2021_2023 = nhanes_data_extraction(url_alc_2021_2023)
```

Here

- demographics extracated dataset into df_demo_2021_2023
- Body Measures extracated dataset into df_bmx_2021_2023
- Diatary data review extracated dataset into df_dr1review_2021_2023
- Laboratory Data: alpha-1-Acid Glycoprotein extracated dataset into df_acid_glycoprotin_2021_2023
- Questionnaire Data: Alcohol Use data load extracated dataset into df_alc_2021_2023

0.5.2 Extracting data 2017-2020 National Helath and Nutrition Examination Center (Nhanes)

```
[6]: # Demographics
df_demo_2017_2020 = nhanes_data_extraction(url_demo_2017_2020)

# Body Measures
df_bmx_2017_2020 = nhanes_data_extraction(url_bmx_2017_2020)

# Diatary data review
df_dr1review_2017_2020 = nhanes_data_extraction(url_dr1review_2017_2020)

#Laboratory Data: alpha-1-Acid Glycoprotein
df_acid_glycoprotin_2017_2020 = □
↳nhanes_data_extraction(url_acid_glycoprotin_2017_2020)

# Questionnaire Data: Alcohol Use data load
df_alc_2017_2020 = nhanes_data_extraction(url_alc_2017_2020)
```

Here

- demographics extracated dataset into df_demo_2017_2020
- Body Measures extracated dataset into df_bmx_2017_2020
- Diatary data review extracated dataset into df_dr1review_2017_2020
- Laboratory Data: alpha-1-Acid Glycoprotein extracated dataset into df_acid_glycoprotin_2017_2020
- Questionnaire Data: Alcohol Use data load extracated dataset into df_alc_2017_2020

0.6 Data Cleaning and Preparation

```
[7]: # Merge demographics with body measures for 2021-2023
df_2021_2023 = pd.merge(df_demo_2021_2023, df_bmx_2021_2023, on='SEQN')
df_2021_2023 = pd.merge(df_2021_2023, df_alc_2021_2023, on='SEQN')

# Merge dietary, AGP, and demographics data
df_2021_2023 = pd.merge(df_2021_2023, df_dr1review_2021_2023, on='SEQN')
df_2021_2023 = pd.merge(df_2021_2023, df_acid_glycoprotin_2021_2023, on='SEQN')

# Merge demographics with body measures for 2017-2020
df_2017_2020 = pd.merge(df_demo_2017_2020, df_bmx_2017_2020, on='SEQN')
df_2017_2020 = pd.merge(df_2017_2020, df_alc_2017_2020, on='SEQN')

# Merge dietary, AGP, and demographics data
df_2017_2020 = pd.merge(df_2017_2020, df_dr1review_2021_2023, on='SEQN')
df_2017_2020 = pd.merge(df_2017_2020, df_acid_glycoprotin_2021_2023, on='SEQN')

# Combine both datasets for comparison 2021-2023 and 2017-2020
df_combined = pd.concat([df_2021_2023, df_2017_2020], ignore_index=True)

# Select relevant columns bmx_acl
df_combined_demo_bmx_acl = df_combined[['RIAGENDR', 'RIDAGEYR', 'BMXBMI',
    ↪ 'ALQ121', 'ALQ151']]

# Rename columns for clarity
df_combined_demo_bmx_acl.rename(columns={
    'RIAGENDR': 'Gender',
    'RIDAGEYR': 'Age',
    'BMXBMI': 'BMI',
    'ALQ121': 'Alcohol_Consumption',
    'ALQ151': 'Alcohol_Frequency'
}, inplace=True)

# Handle missing values
df_combined_demo_bmx_acl.dropna(inplace=True)
df_combined_demo_bmx_acl
```

```
[7]:
```

	Gender	Age	BMI	Alcohol_Consumption	Alcohol_Frequency
0	2.0	44.0	29.7	10.0	2.0
1	2.0	44.0	29.7	10.0	2.0
2	2.0	44.0	29.7	10.0	2.0
3	2.0	44.0	29.7	10.0	2.0
4	2.0	44.0	29.7	10.0	2.0
...	...	...	...	...	...
15235	2.0	46.0	32.6	6.0	2.0
15236	2.0	46.0	32.6	6.0	2.0
15237	2.0	46.0	32.6	6.0	2.0
15238	2.0	46.0	32.6	6.0	2.0
15239	2.0	46.0	32.6	6.0	2.0

[12346 rows x 5 columns]

- dataframe df_2021_2023 Merge demographics with body measures for 2021-2023
- dataframe df_2021_2023 Merge dietary, AGP, and demographics data
- dataframe df_2017_2020 Merge demographics with body measures for 2017-2020
- dataframe df_2017_2020 Merge dietary, AGP, and demographics data
- df_combined dataframe Combine both datasets for comparison 2021-2023 and 2017-2020
- Select relevant columns bmx_acl
- Rename columns for clarity
- Handle missing values

```
[8]: # Select relevant columns dietary and agp
df_diet_agp = df_combined[['RIAGENDR', 'RIDAGEYR', 'DR1IKCAL', 'DR1IPROT', 'DR1ICARB', 'DR1ISUGR', 'DR1ITFAT', 'LBXAGP']]

# Rename columns for clarity
df_diet_agp.rename(columns={
    'RIAGENDR': 'Gender',
    'RIDAGEYR': 'Age',
    'DR1IKCAL': 'Calories',
    'DR1IPROT': 'Protein',
    'DR1ICARB': 'Carbohydrates',
    'DR1ISUGR': 'Sugars',
    'DR1ITFAT': 'Total_Fat',
    'LBXAGP': 'AGP'
}, inplace=True)

#Handle missing values
df_diet_agp.dropna(inplace=True)
df_diet_agp
```

```
[8]:
```

	Gender	Age	Calories	Protein	Carbohydrates	Sugars	Total_Fat	\
0	2.0	44.0	6.0	0.20	1.23	0.00	0.01	
1	2.0	44.0	7.0	0.52	0.79	0.76	0.15	

2	2.0	44.0	308.0	10.70	55.11	0.81	5.31
3	2.0	44.0	61.0	0.47	14.31	9.95	0.18
4	2.0	44.0	0.0	0.00	0.00	0.00	0.00
...	...	...	...	...	...	...	...
15246	2.0	49.0	146.0	7.32	14.11	3.92	8.68
15247	2.0	49.0	40.0	0.00	9.62	6.83	0.12
15248	2.0	49.0	248.0	3.53	45.14	17.87	5.95
15249	2.0	49.0	252.0	15.35	15.60	0.60	14.26
15250	2.0	49.0	4.0	0.00	0.47	0.00	0.00

	AGP
0	1.010
1	1.010
2	1.010
3	1.010
4	1.010
...	...
15246	0.745
15247	0.745
15248	0.745
15249	0.745
15250	0.745

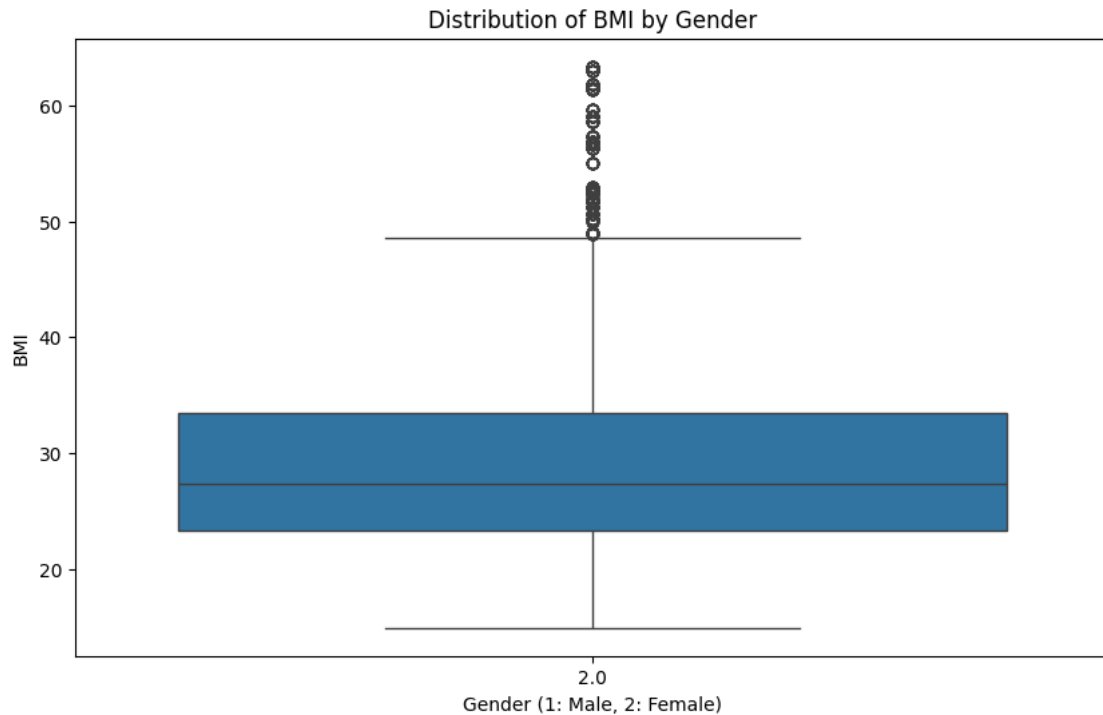
[14106 rows x 8 columns]

- Select relevant columns diary food and apg consumsome
- Rename columns for clarity and relevent column define
- Handle missing values

0.7 Create Visualizations

0.7.1 Distribution of BMI by Gender

```
[9]: plt.figure(figsize=(10, 6))
sns.boxplot(x='Gender', y='BMI', data=df_combined_demo_bmx_acl)
plt.title('Distribution of BMI by Gender')
plt.xlabel('Gender (1: Male, 2: Female)')
plt.ylabel('BMI')
plt.show()
```

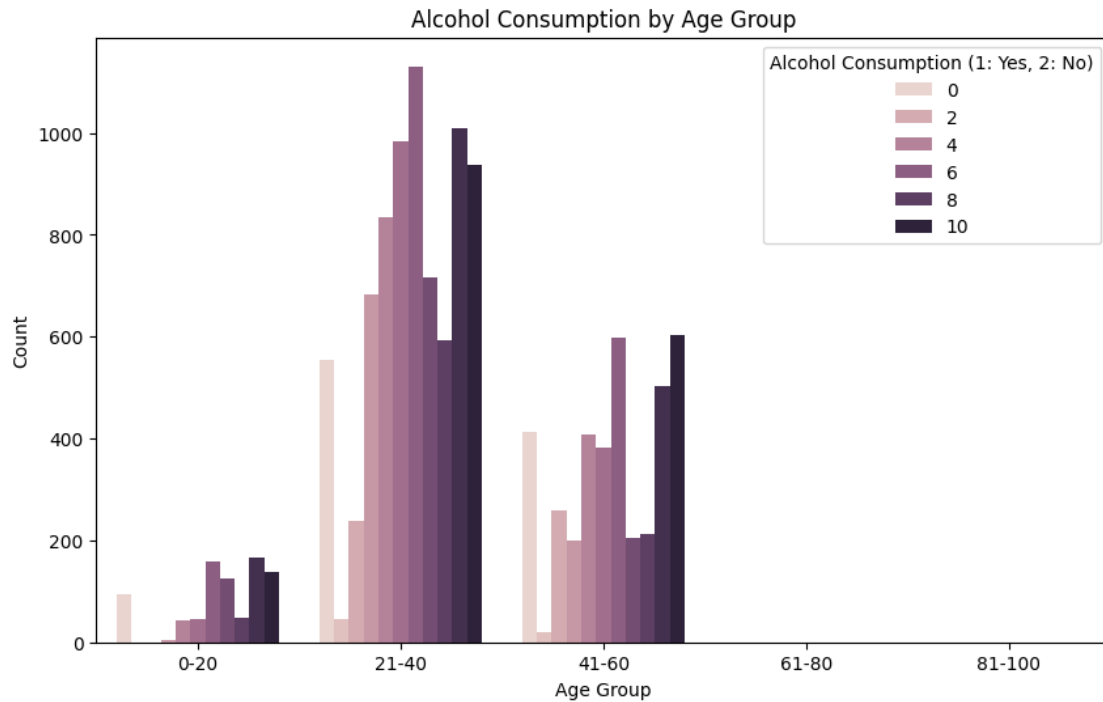


Insight:

This plot shows the distribution of BMI across genders. It can help identify if there are significant differences in BMI between males and females.

0.7.2 Alcohol Consumption by Age Group

```
[10]: df_combined_demo_bmx_acl['Age_Group'] = pd.cut(df_combined_demo_bmx_acl['Age'],
    ↪ bins=[0, 20, 40, 60, 80, 100], labels=['0-20', '21-40', '41-60', '61-80',
    ↪ '81-100'])
plt.figure(figsize=(10, 6))
sns.countplot(x='Age_Group', hue='Alcohol_Consumption',
    ↪ data=df_combined_demo_bmx_acl)
plt.title('Alcohol Consumption by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Count')
plt.legend(title='Alcohol Consumption (1: Yes, 2: No)')
plt.show()
```

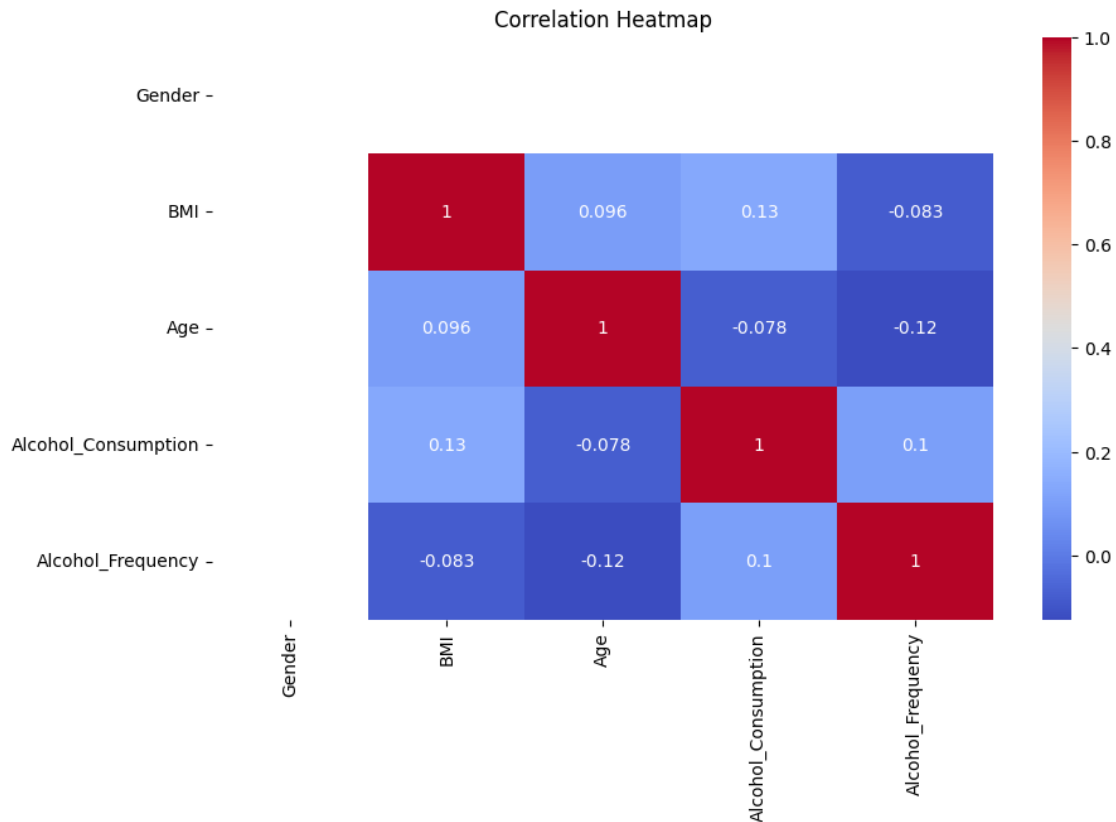



Insight:

This plot shows the prevalence of alcohol consumption across different age groups. It can help identify trends in alcohol use by age.

Correlation Heatmap of BMI and alcohol consumption

```
[11]: plt.figure(figsize=(10, 6))
bmi_alc = 
↳ df_combined_demo_bmx_acl[['Gender', 'BMI', 'Age', 'Alcohol_Consumption', 'Alcohol_Frequency']]
sns.heatmap(bmi_alc.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

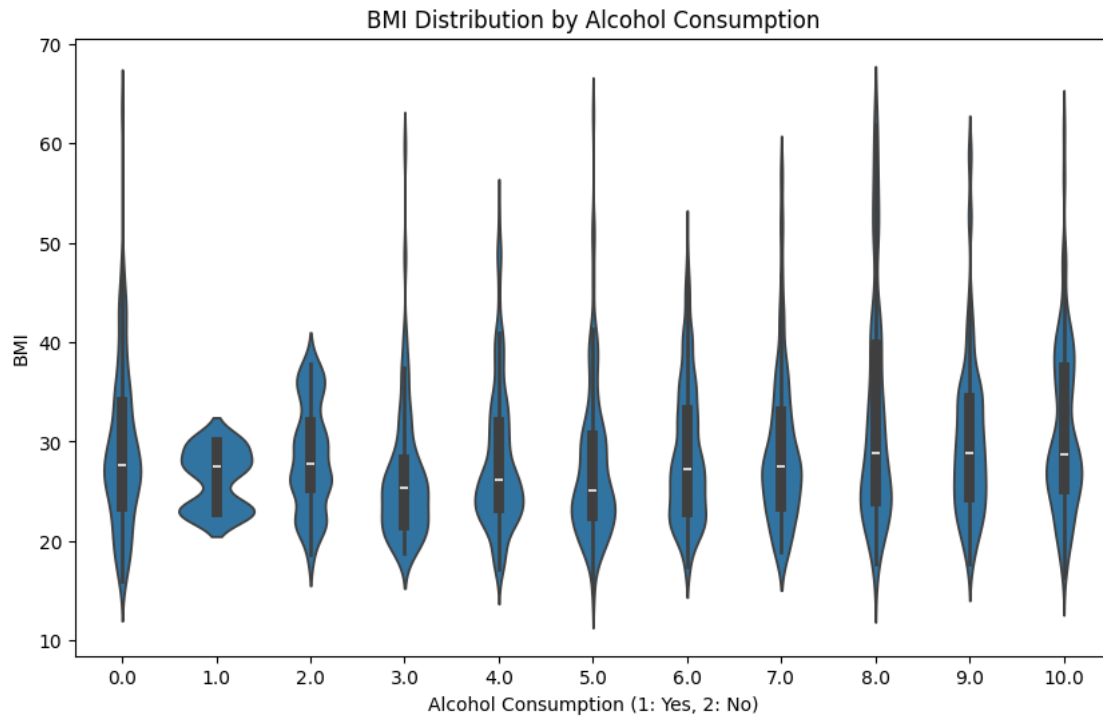


Insight:

This heatmap shows the correlation between different variables (BMI, Age, Alcohol Consumption). It can help identify relationships between variables. AG-Group is Categorical data we avoid here

0.7.3 BMI Distribution by Alcohol Consumption

```
[12]: plt.figure(figsize=(10, 6))
sns.violinplot(x='Alcohol_Consumption', y='BMI', data=df_combined_demo_bmx_acl)
plt.title('BMI Distribution by Alcohol Consumption')
plt.xlabel('Alcohol Consumption (1: Yes, 2: No)')
plt.ylabel('BMI')
plt.show()
```

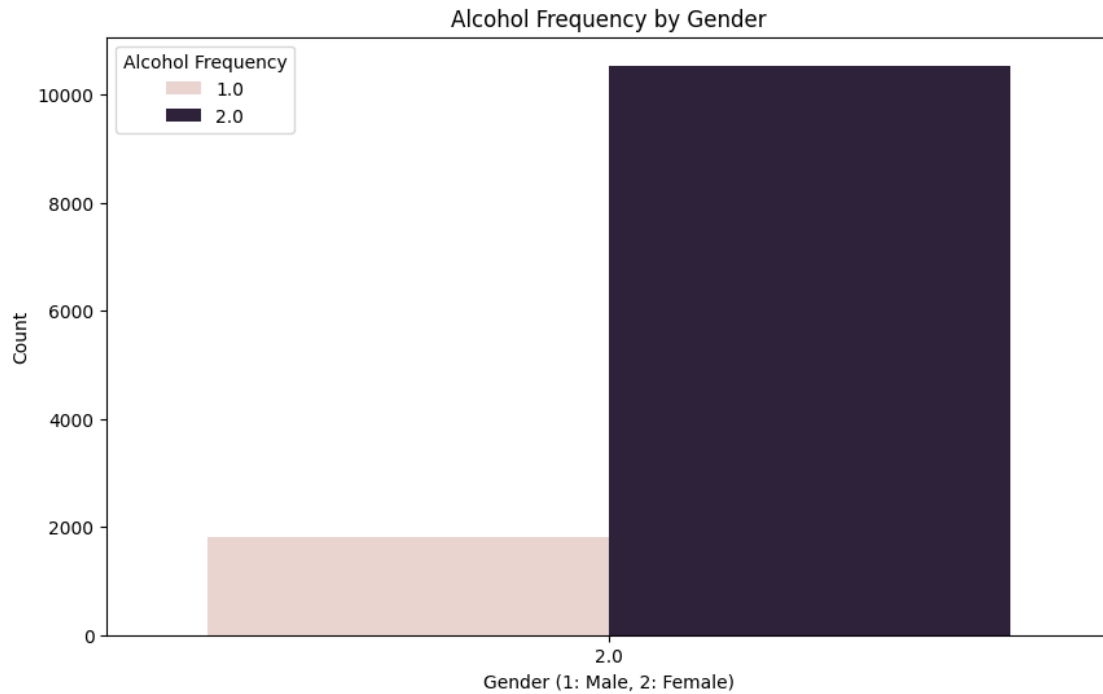


Insight:

This plot shows the distribution of BMI for individuals who consume alcohol versus those who do not. It can help identify if alcohol consumption is associated with differences in BMI.

0.7.4 Alcohol Frequency by Gender

```
[13]: plt.figure(figsize=(10, 6))
sns.countplot(x='Gender', hue='Alcohol_Frequency',
data=df_combined_demo_bmx_acl)
plt.title('Alcohol Frequency by Gender')
plt.xlabel('Gender (1: Male, 2: Female)')
plt.ylabel('Count')
plt.legend(title='Alcohol Frequency')
plt.show()
```

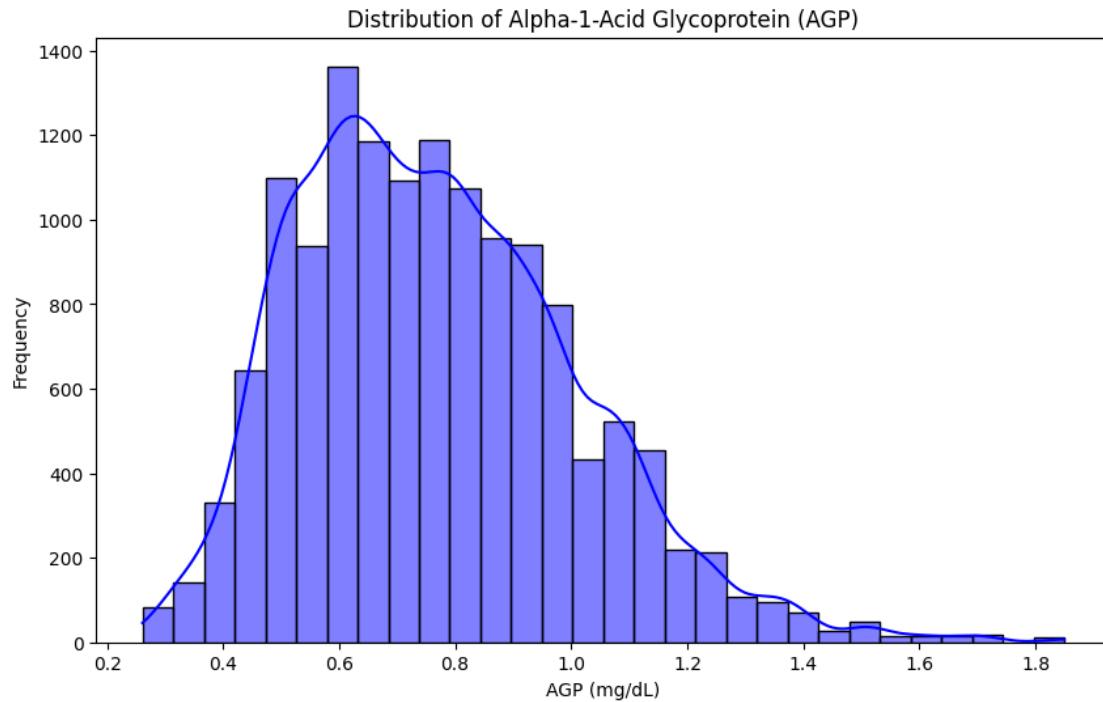


Insight:

This plot shows the frequency of alcohol consumption by gender. It can help identify if there are differences in drinking patterns between males and females.

0.7.5 Distribution of Alpha-1-Acid Glycoprotein (AGP)

```
[14]: plt.figure(figsize=(10, 6))
sns.histplot(df_diet_agp['AGP'], bins=30, kde=True, color='blue')
plt.title('Distribution of Alpha-1-Acid Glycoprotein (AGP)')
plt.xlabel('AGP (mg/dL)')
plt.ylabel('Frequency')
plt.show()
```

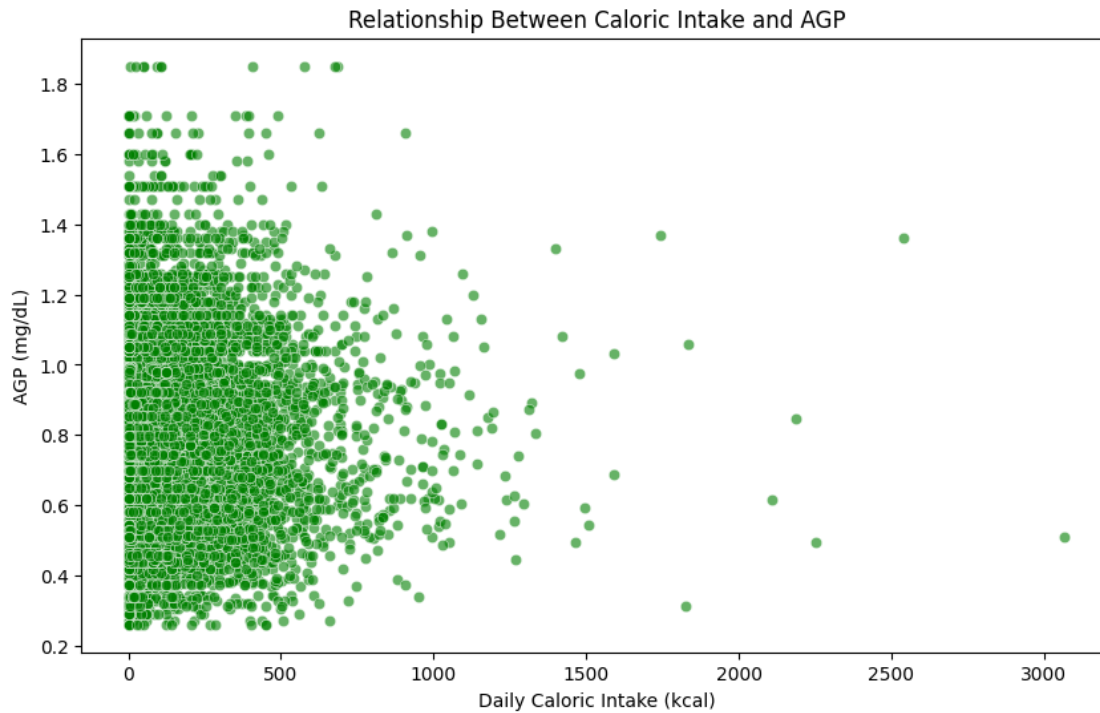


Insight:

This plot shows the distribution of AGP levels in the population. AGP is a marker of inflammation, and higher levels may indicate chronic inflammation or immune response. The distribution can help identify if there are outliers or a skewed distribution.

0.7.6 Relationship Between Caloric Intake and AGP

```
[15]: plt.figure(figsize=(10, 6))
sns.scatterplot(x='Calories', y='AGP', data=df_diet_agp, alpha=0.6,
               color='green')
plt.title('Relationship Between Caloric Intake and AGP')
plt.xlabel('Daily Caloric Intake (kcal)')
plt.ylabel('AGP (mg/dL)')
plt.show()
```



Insight:

This scatterplot explores whether there is a relationship between daily caloric intake and AGP levels. Higher caloric intake might be associated with higher inflammation markers, but this would require further statistical analysis to confirm.

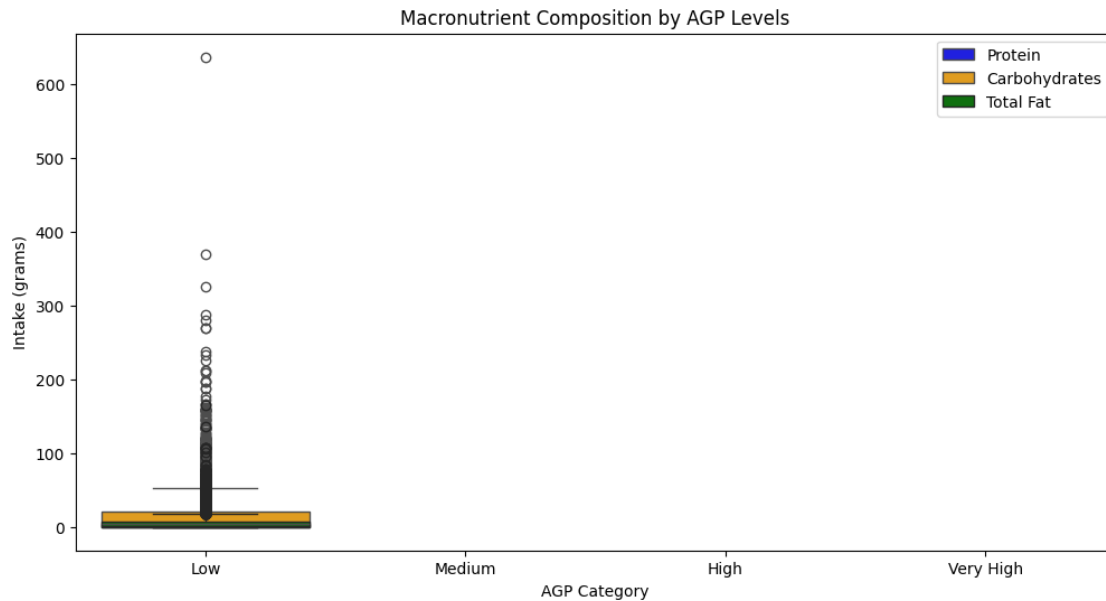
0.7.7 Macronutrient Composition by AGP Levels

We can group individuals by AGP levels (low, medium, high) and compare their macronutrient intake (protein, carbohydrates, fats).

```
[16]: # Create AGP categories
df_diet_agp['AGP_Category'] = pd.cut(df_diet_agp['AGP'], bins=[0, 50, 100, 150, 200], labels=['Low', 'Medium', 'High', 'Very High'])

# Plot macronutrient composition by AGP category
plt.figure(figsize=(12, 6))
sns.boxplot(x='AGP_Category', y='Protein', data=df_diet_agp, color='blue', label='Protein')
sns.boxplot(x='AGP_Category', y='Carbohydrates', data=df_diet_agp, color='orange', label='Carbohydrates')
sns.boxplot(x='AGP_Category', y='Total_Fat', data=df_diet_agp, color='green', label='Total Fat')
plt.title('Macronutrient Composition by AGP Levels')
plt.xlabel('AGP Category')
```

```
plt.ylabel('Intake (grams)')
plt.legend()
plt.show()
```

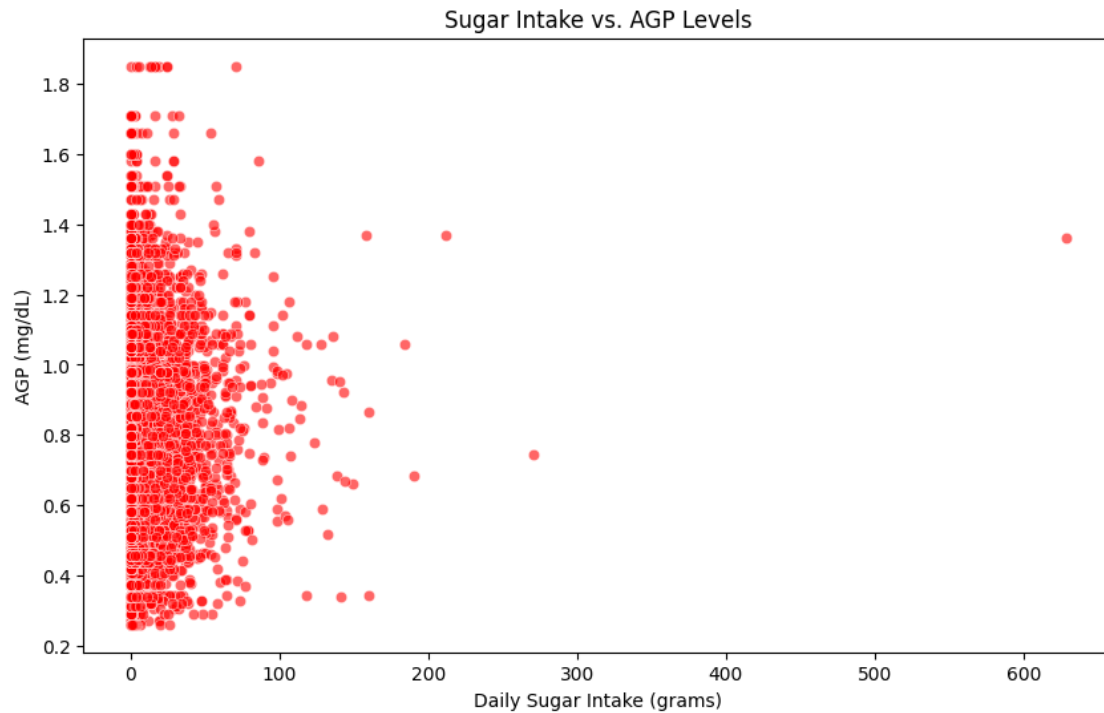


Insight:

This plot compares the intake of protein, carbohydrates, and fats across different AGP levels. It can help identify if certain macronutrient patterns are associated with higher or lower inflammation markers.

0.7.8 Sugar Intake vs. AGP Levels

```
[17]: plt.figure(figsize=(10, 6))
sns.scatterplot(x='Sugars', y='AGP', data=df_diet_agp, alpha=0.6, color='red')
plt.title('Sugar Intake vs. AGP Levels')
plt.xlabel('Daily Sugar Intake (grams)')
plt.ylabel('AGP (mg/dL)')
plt.show()
```

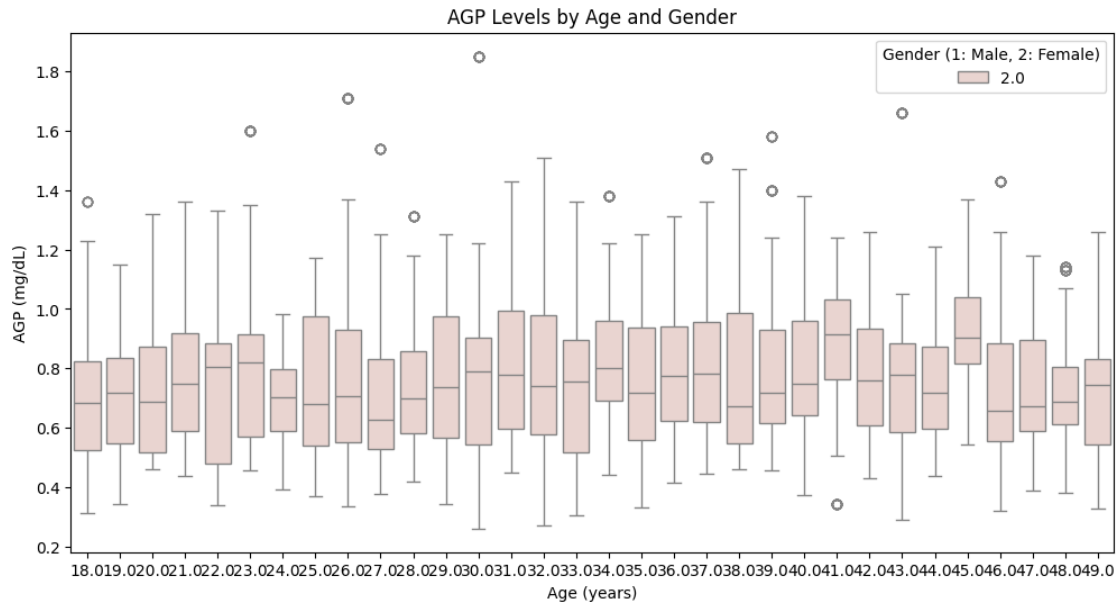


Insight:

This scatterplot explores the relationship between sugar intake and AGP levels. High sugar intake is often linked to inflammation, so this visualization can help identify if there is a trend.

0.7.9 AGP Levels by Age and Gender

```
[18]: plt.figure(figsize=(12, 6))
sns.boxplot(x='Age', y='AGP', hue='Gender', data=df_diet_agp)
plt.title('AGP Levels by Age and Gender')
plt.xlabel('Age (years)')
plt.ylabel('AGP (mg/dL)')
plt.legend(title='Gender (1: Male, 2: Female)')
plt.show()
```

Insight:

This plot shows how AGP levels vary by age and gender. It can help identify if certain age groups or genders are more prone to higher inflammation markers.

0.8 Based on the visualizations, Here is data insights:

BMI Distribution by Gender: There may be differences in BMI between males and females, which could be further investigated.

Alcohol Consumption by Age Group: Younger age groups may have higher rates of alcohol consumption compared to older age groups.

Correlation Heatmap: There may be correlations between variables like BMI and alcohol consumption, which could be explored further.

BMI Distribution by Alcohol Consumption: Individuals who consume alcohol may have different BMI distributions compared to those who do not.

Alcohol Frequency by Gender: Males and females may have different patterns of alcohol consumption frequency.

AGP Distribution: The distribution of AGP levels can help identify populations at risk of chronic inflammation.

Caloric Intake and AGP: There may be a relationship between high caloric intake and elevated AGP levels, suggesting a link between overeating and inflammation.

Macronutrient Composition: Differences in macronutrient intake (e.g., high fat or high carbohydrate diets) may be associated with varying AGP levels.

Sugar Intake and AGP: High sugar intake may correlate with higher AGP levels, reinforcing the link between diet and inflammation.

Age and Gender Differences: AGP levels may vary by age and gender, indicating that certain demographics are more susceptible to inflammation.

[]: